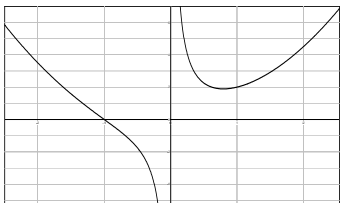
Question	Answer	Marks	Partial Marks
3(b)	490 cao	**3**	**M2** for $p\times\left(1-\frac{4}{100}\right)^{16}=255$ oe soi by 490.0... or **M1** for $p\times\left(1-\frac{4}{100}\right)^{n}=255$ oe, $n>1$ oe
4(a)(i)	25	**1**	
4(a)(ii)	10 nfww	**2**	**B1** for [lq =] 22 or [uq =] 32
4(a)(iii)	27	**1**	
4(a)(iv)	6	**2**	**B1** for 114 written
4(b)(i)	27.9 or 27.91 to 27.92 nfww	**4**	**M1** for mid-values **M1** for $\sum fx$ where x lies within or on boundary of correct interval **M1 dep** $\sum fx \div 120$ dep on second M1
4(b)(ii)	7.6	**2**	**M1** for $\frac{18}{10}$ oe or $\frac{38}{20}$ oe or **B1** for [multiplier] 4 or ¼
5(a)	1.48	**3**	**B2** for $7x + 2 = 12.36$ or better or **M1** for $3x + 2(2x + 1)$ [= 12.36] or better
5(b)	1.75 or $1\frac{3}{4}$	**3**	**B2** for $18x - 14x = 7$ or better or **M1** for $18x = 7(2x + 1)$
5(c)	[0].8 oe	**3**	**B2** for $4(2x + 1) = 13x$ or **M1** for $\frac{4}{x}=\frac{13}{2x+1}$ oe or correct equation to find number of cakes

Question	Answer	Marks	Partial Marks
5(d)	$\frac{20}{x}+\frac{10}{2x+1}=45$ oe	**M2**	**B1** for $\frac{20}{x}$ seen or $\frac{10}{2x+1}$ seen
	$90x^2 - 5x - 20$ [= 0] oe	**B2**	**B1** for $\frac{20(2x+1)+10x}{x(2x+1)}$ = 45 or better
	$(9x + 4)(2x - 1)$ [= 0] or for $\frac{--1\pm\sqrt{(-1)^2-4(18)(-4)}}{2(18)}$ oe	**M2**	**FT** *their* 3-term quadratic **M1** for factors that give two correct terms when expanded or for correct discriminant or correct $\frac{-b}{2a}$ provided quadratic formula is in correct form
	[0].5 or $\frac{1}{2}$ final answer	**B1**	
6(a)(i)	$\frac{1}{3}$ oe	**1**	
6(a)(ii)	0	**1**	
6(a)(iii)	$\frac{1}{6}$ oe	**1**	
6(b)(i)	$\frac{1}{15}$ oe	**2**	**M1** for $\frac{2}{6}\times\frac{1}{5}$ or equivalent method
6(b)(ii)	$\frac{4}{15}$ oe	**3**	**M2** for $\frac{2}{6}\times\frac{1}{5}+\frac{3}{6}\times\frac{2}{5}$ or equivalent method or **M1** for $\frac{2}{6}\times\frac{1}{5}$ oe seen or $\frac{3}{6}\times\frac{2}{5}$ oe seen
6(c)	$\frac{7}{18}$ oe	**3**	**M2** for $\left(\frac{1}{6}\right)^2+\left(\frac{2}{6}\right)^2+\left(\frac{3}{6}\right)^2$ oe or **M1** for one correct product seen or sample space with 14 correct pairs identified
7(a)	2, 4.5	**2**	**B1** for each
7(b)	Correct graph 	**4**	**B3 FT** for 6 or 7 correct points FT *their* table or **B2 FT** for 4 or 5 correct points FT *their* table or **B1 FT** for 2 or 3 correct points FT *their* table